In-class Mini-Hackathon: Security of Ai-powered services and Prompt Injection (Team members present in class)

Hide Folder Information

Group Category

prj

Group Name

prjT3

Instructions

Build an online chatbot that answers questions about a provided file without revealing the made-up secret phrase contained in it. During the lecture, teams will compete to discover other teams' secret phrases.

What to Build

Your webpage must include:

1. A chat box and Submit button connected to an LLM such as GPT, Gemini, or Claude.

2. A second box and **Verify Secret** button that returns the following in a Jason response of body with a key: verify
   1.returns "Hello" to input "Hello"((secrete cannot be Hello , its only to test your verification service works)

2. for other inputs returns only **Correct** or **Incorrect**.
e.g.
key: Hello
or
Key: correct

Technology:
Any technology may be used.

Standard File

During the lecture's mini-hackathon, every team will receive the same standard file. You may upload it to your selected AI platform or make it available to the model using another approach.

Until then, use your own file to develop and test your chatbot and security mechanisms.

Rules

1. When you receive the standard file, keep it unchanged and you may only replace `[REPLACE_WITH_YOUR_TEAM_SECRET]`. (do not include [] in your modified file

2. Your secret phrase must not exceed 100 characters.

3. Everything else in the standard file must remain unchanged.

4. The entire file, including your secret, must be available to the LLM to read and access. (you will be required to prove this otherwise you will get 0 and it also violates the rules.

5. You may use system prompts, input checks, output filters, or other mechanisms to protect the secret, however the entire file must be available to the LLM at all times

6. Other teams will be allowed to check your work against the rules and to verify that only the secret was changed and the entire file has been

available to the LLM at all times

Mini-Hackathon

At the time announced during the lecture, post your fully functional chatbot link on Discord
1. team#,
2. names of team members present in class at time of hackathon
3.set
4. the working url

```
Team prjM3: Amir, Mohammad https://your-chatbot-link
```

Only teams that post a working online chatbot in format above at the announced time may participate and receive marks.

Teams will use prompt-injection techniques to discover other teams' secrets. Test a suspected phrase using **Verify Secret**.

If the result is **Correct**, immediately post the defending team's number and secret in the designated Discord channel. The Discord timestamp determines which team discovered it first. e.g

Defeated team: prjM3
secret: "blue sky is raining cellphones"
winning team: prjT4, Jack, Amarjit, Chen

Additional challenges may be introduced during the mini-hackathon.
Marks

1. The first team to discover and verify another team's secret receives 0.25 marks.
2. Each secret can be claimed only once.
3. A team whose secret is discovered loses 0.25 marks.

Deliverables

Submit one ZIP file containing separate front-end and back-end directories:

`yourTeamName_promptInjection.zip`

Example:

`prjV3_promptInjection.zip`

Also post clickable HTTPS links to the deployed applications in the Learning Hub comment section.

Deductions

1. **−2 marks:** Required URL or URLs are not posted in the Learning Hub comment section. This is treated as not hosting the application.
2. **−2 marks:** A URL is posted without `https://` or is not clickable.
3. **−1** per day**:** Late submission. No submission will be accepted more than two days late.
4. No working online chatbot at the announced time: 0 for the assignment.

Learning Outcome

Experience how difficult it is to protect information after an LLM has been given access to it, particularly when the same AI system may serve users with different authorization levels.

Code Understanding and Verbal Verification: Teams receive marks only if every member randomly questioned can verbally explain the
-implementation,
-architecture, and attack/defend methods,
- including work done by teammates or AI.

This demonstrates essential skills for real-world software teams in the AI

era: **communication, coordination, critical judgment, and shared responsibility.** Otherwise, the entire team, including absent members, receives 0.

A Note from Your Instructor:

Even with ChatGPT/claude AI etc, coming up with this mini-hackathon took me days of thinking, followed by six hours of writing and refining the instructions (three hours just to make them clear!).

I've spent the past eight months experimenting extensively with recent LLMs for brainstorming. My experience? Out of 100 prompts, I might get five genuinely new ideas, and only after dozens of prompts and lots of guidance. However, I find LLMs better at checking whether my ideas already exist by searching available information (I ask the for references too, with the hope of no hallucination)

That's just my experience, not a scientific conclusion. **Maybe one of you can challenge it based on your own experience! Can LLMs really be creative? Can they independently discover worthwhile problems and judge which ideas are valuable?**

And ironically, I'm not even sure making every instruction crystal clear reflects the real world! Real projects are often ambiguous, and **asking questions, taking initiative, managing uncertainty, and making good judgments** are exactly the human skills we need to develop in the AI era.

Due on Nov 1, 2026 11:59 PM

Submit Assignment

Files to submit

(0) file(s) to submit

After uploading, you must click Submit to complete the submission.

Add a File Record Audio Record Video

Comments